MATHS MANTHAN
• TUITION CLASSES •
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**Subject:** Mathematics | **CBSE CLASS 10 – TEST (SET – B)** | **Time:** 1 Hour
**Class:** Class X (CBSE) | **Chapter 5: Arithmetic Progressions (AP)** | **MM: 30**

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory.
2. The question paper consists of 13 questions divided into three Sections: A, B, and C.
3. **Section A** comprises 5 questions of 1 mark each (3 MCQs and 2 Assertion-Reason).
4. **Section B** comprises 5 questions of 2 marks each (Short Answer Type).
5. **Section C** comprises 3 questions of 5 marks each (Long Answer Type).
6. Use of calculator is not permitted.

## SECTION A: Multiple Choice Questions (1 Mark Each) (5 × 1 = 5 Marks)

**Q1.** If the first three terms of an A.P. are $3p-1, 3p+5, 5p+1$ respectively, then the value of $p$ is: [1]

(A) 4 (B) 5 (C) 2 (D) 3

**Q2.** The next term of the A.P. $\sqrt{7}, \sqrt{28}, \sqrt{63}, \ldots$ is: [1]

(A) $\sqrt{70}$ (B) $\sqrt{84}$ (C) $\sqrt{98}$ (D) $\sqrt{112}$

**Q3.** The common difference of an A.P. in which $a_{20} - a_{15} = 20$, is: [1]

(A) 4 (B) 5 (C) 6 (D) 3

**Q4.** **Assertion (A):** Common difference of the A.P. $5, 1, -3, -7, \ldots$ is 4.
**Reason (R):** Common difference of an A.P. $a_1, a_2, a_3, \ldots, a_n$ is obtained by $d = a_n - a_{n-1}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

**Q5.** **Assertion (A):** The eighth term of the A.P. $\frac{1}{m}, \frac{1+2m}{m}, \frac{1+4m}{m}, \ldots$ is $\frac{1+14m}{m}$.
**Reason (R):** The $n^{\text{th}}$ term of an A.P. is given by $a_n = a + (n-1)d$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

## SECTION B: Short Answer Type Questions (2 Marks Each) (5 × 2 = 10 Marks)

**Q6.** Find the $11^{\text{th}}$ term from the last term of the A.P.: $10, 7, 4, \ldots, -62$. [2]

**Q7.** Find the number of natural numbers between 101 and 999 which are divisible by both 2 and 5. [2]

**Q8.** Check whether $-150$ is a term of the A.P. $11, 8, 5, 2, \ldots$ [2]

**Q9.** How many terms of the A.P. $27, 24, 21, \ldots$ should be taken so that their sum is zero? **[2]**

**Q10.** In an A.P., if $S_n = 3n^2 + 4n$, find the $r^{\text{th}}$ term of the A.P. **[2]**

## SECTION C: Long Answer Type Questions (5 Marks Each) (3 × 5 = 15 Marks)

**Q11.** The sum of four consecutive numbers in an A.P. is 32 and the ratio of the product of the first and the last term to the product of two middle terms is 7 : 15. Find the numbers. **[5]**

**Q12.** How many terms of the A.P. $9, 17, 25, \ldots$ must be taken to give a sum of 636? **[5]**

**Q13.** A manufacturer of TV sets produced 600 sets in the third year and 700 sets in the seventh year. Assuming that the production increases uniformly by a fixed number every year, find: **[5]**

(i) The production in the first year.

(ii) The production in the $10^{\text{th}}$ year.

(iii) The total production in first 7 years.

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★